

A Comparative Study of Classical and Transformer-Based Models for Phishing Email Detection

Bachelor Semester Project 6

João Bernardo SOUSA
FARIA

Student ID: 0232421751
University of Luxembourg
joao.sousa.003@student.uni.lu

Secondary Language:
French

Salima LAMSIYAH
Department of Computer Science
University of Luxembourg
salima.lamsiyah@uni.lu

ABSTRACT

The project is developed by student **João Bernardo Sousa Faria**, under the supervision of their tutor **Salima Lamsiyah**. The project aims to detect phishing emails with the use of Natural Language Processing (NLP) techniques. Phishing attacks are a common yet dangerous threat in cybersecurity, which results in users exposing sensitive information. With this, phishing email detection is essential for improving security in digital environments.

The system treats phishing detection as a **binary text classification problem** and analyzes the content of emails to determine whether a message is legitimate or malicious. To achieve this goal, the project compares classical machine learning approaches based on TF-IDF features and classifiers such as **Logistic Regression** with transformer-based models such as **BERT**.

The objective of this project is to **evaluate and compare the performance of both approaches**, with the goal of recognizing their respective strengths and weaknesses when it comes to detecting phishing emails.

Overview of the Scientific Section of the Project

The project investigates the question “*How do classical feature-based machine learning models compare with transformer-based models in phishing email detection tasks?*”.

The objective of the study is to **understand different NLP techniques which are used for text classification, as well as phishing detection**. The study compares classical machine learning methods and transformer-based architectures based on factors such as **model complexity** and **performance**. Additionally, the study highlights conditions in which each model thrives.

Overview of the Technical Section of the Project

The technical section of the project focuses on implementing and evaluating a phishing email detection system. The tools for developing the system are as follows:

- **Main Programming Language:** Python;

- **Data Processing and Preparation:** libraries such as NumPy and Pandas;
- **Classical machine learning models:** implemented with Scikit-learn, with techniques such as TF-IDF vectorization, combined with classifiers such as Logistic Regression;
- **Transformer-based models:** implemented with PyTorch together with the HuggingFace Transformers Library, allowing fine-tuning of pretrained models such as BERT for phishing email classification
- **Dataset:** publicly available dataset containing emails labeled as “legitimate” or “malicious”;
- **Performance:** evaluated using metrics such as accuracy and F1-score.
- **Visualization:** using libraries such as Matplotlib for illustrating experimental results, as well as studying them.